

Gaming ISO 20022 Payment Messages: An Adversarial Data-Quality Benchmark for Purpose Codes, Party Fields, and Message Paths

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Abstract—ISO 20022 gives payment controls more structured data, but message creators still decide where names, addresses, and purpose information appear. We study *message gaming*: deliberate edits to purpose codes, party fields, or the message path that lower a modelled alert probability while preserving the payment’s business meaning. Purpose-code gaming is the best-known case; the benchmark also covers misfielding and message-path changes, eleven manipulations in all. A seeded generator creates synthetic pain.001 and pacs.008 records, adds ordinary data-quality defects at a controlled rate, and labels adversarial edits. The evaluation is synthetic and relative to a published screening abstraction, not to any institution’s controls. Each of three 200,000-record runs contains about 5.0% gamed messages among benign, defective, and unevaded illicit ones. We also introduce screening-evasion lift (SEL), which measures how much a manipulation increases modelled pass-through. Seven detectors are compared, including two standard reference methods, with every learned model cross-fitted so that no record is scored by a model that trained on it. Under two-fold cross-fitting, the supervised gradient-boosted model reaches AUC 0.883 and recovers 16.8% of gamed messages at an exact 1% alert capacity, against a ceiling of 20.2%. It then falls from AUC 0.769 to 0.377 on a misfielding family withheld from training, because that family’s signature is benign-only in the training data. The label-free detectors reach AUC 0.750 to 0.777 but recover only 3.8% to 10.1% at the same capacity, and their ordering by AUC is the reverse of their ordering by recovery. SEL ranges from 1.33 to 1.79 in the main calibration. Detection difficulty does not follow the P1/P2/P3 action taxonomy; it follows whether an edit leaves a contradiction inside the message.

Index Terms—ISO 20022, payment data quality, purpose codes, sanctions screening, anti-money laundering, adversarial evaluation, synthetic benchmark

I. INTRODUCTION

A payment message is both an instruction and a compliance record. Names, addresses, amounts, and purpose information may feed sanctions, fraud, and anti-money-laundering controls. The move to ISO 20022 makes much of that information easier to parse. In CBPR+, purely unstructured party and agent addresses cease to be supported from 14 November 2026; town and country must at least appear in designated fields [1]. CPMI’s harmonised data requirements similarly encourage consistent structured data and registered codes by end-2027, although CPMI is explicit that the requirements are guidance rather than a regulatory mandate [2].

Structure does not remove discretion. A message creator still chooses which optional fields to populate, which purpose

code to declare, how much detail to place in free text, and sometimes which initiation or routing path is used. Screening is also selective. Published Swift guidance identifies elements that normally warrant direct screening and allows other fields to provide context instead [3]. Purpose and category-purpose codes may influence processing or monitoring, yet market-practice guidance notes that these codes are often misunderstood or placed in the wrong field [4]. An ordinary data-quality error and a deliberate low-scrutiny choice can therefore leave similar evidence.

That overlap creates a practical evaluation problem. A detector can look good by finding malformed records, even though malformed records are mostly benign. It can also earn a high AUC by learning the edit signatures of one generator, then fail on a manipulation that was absent from training. The relevant question is narrower than “can the model spot unusual data?” It is whether deliberate changes remain distinguishable from routine error at an alert capacity a payment operation could actually use.

The work contributes eleven message-level manipulations with explicit capability assumptions and costs, and a seeded synthetic benchmark with a tunable ordinary-error floor, a transparent screening abstraction, and leave-one-family-out testing. It introduces screening-evasion lift (SEL), which separates an adversary’s gain from the portion removed by a detector. It compares seven detectors, two of them standard reference methods, under cross-fitting and exact alert capacities with confidence intervals. The central empirical result is that the taxonomy explains what the adversary changes, but not what a detector can see. Detectability follows message-internal contradiction rather than the P1/P2/P3 family label. The detectors are conventional on purpose; the benchmark, the operational measurement, and the failure cases are the contribution. No real payment, customer, or institution-specific alert data is used.

II. BACKGROUND AND RELATED WORK

A. Fields and control boundaries

The benchmark covers customer credit transfers at two stages: pain.001 at initiation and pacs.008 between financial institutions. Relevant content includes Purp/Cd, CtgyPurp/Cd, party names and postal addresses, UltmtDbtr, and structured or

unstructured remittance information. Purpose describes the underlying reason for a transaction; category purpose is a higher-level instruction that may affect processing. Their exact use is corridor- and institution-specific, which is why the paper does not assign a universal risk meaning to any ISO code.

The address transition is useful context but not the attack itself. A hybrid address can retain free-text lines while placing town and country in their designated fields. That improves machine readability, but it does not guarantee that the most informative identity or location evidence sits in a field read by every control. Likewise, a valid purpose code can still be implausible for the creditor’s industry or inconsistent with the remittance narrative. The benchmark treats these as consistency questions, not schema errors.

Selective screening is unavoidable. Reading every string as a party name would create an unmanageable false-positive burden, while ignoring contextual fields loses useful evidence. The tested controls therefore represent three layers: structural validation, consistency checks within one message, and comparison with a synthetic transaction history. Message-level checks can catch contradictions, but they cannot recover context that has been removed or shifted to a different path.

B. Related benchmarks and adversarial framing

Public AML resources operate at the transaction or graph level. The Elliptic data set labels Bitcoin transactions [5], AMLSim simulates laundering patterns over account graphs [6], AMLworld models realistic transaction networks with laundering labels [7], and SAML-D supplies a labelled synthetic monitoring dataset [8]. The more recent AMLgentex and Tide generators add partial observability, temporal dynamics, and configurable illicit prevalence [9], [10]. Strategic simulations let laundering agents adapt against a detector [11], and directed-multigraph neural networks have been built to read these graphs [12]. All of these resources study who pays whom, how much, and when. They do not encode the internal placement of ISO 20022 fields, so they cannot express a manipulation whose mechanism is that a name, code, or narrative appears in a different element. Transaction-purpose classification addresses enrichment of missing or noisy labels, rather than a label selected strategically [13]. On the data-quality side, Hammami *et al.* show how noisy free-text party addresses in real payment traffic degrade parsing even as ISO 20022 adds structure, and release an augmented benchmark for it [21]; that is the ordinary-error half of the overlap studied here.

Adversarial machine learning on tabular financial data is the closest methodological line. Ballet *et al.* formalise imperceptibility for tabular perturbations [14], and Cartella *et al.* adapt gradient attacks to imbalanced fraud data [15]. Simonetto *et al.* generate feasible examples under domain constraints, combine gradient and search attacks that respect them, and benchmark tabular robustness across finance, healthcare, and security tasks [16]–[18]. Fok *et al.* show that gradient-based perturbations transfer to tree ensembles in credit-card fraud detection [19]. FraudBench stress-tests policy-grounded banking agents against an adaptive social-engineering adversary [20];

TABLE I
SCOPE OF ADJACENT BENCHMARKS AS STATED BY THEIR AUTHORS. AML SETS: ELLIPTIC, AMLSIM, AMLWORLD, SAML-D, AMLGENTEX, TIDE [5]–[10]. TABULAR ROBUSTNESS: [14]–[18]. FRAUDBENCH: [20].

Dimension	AML sets	Tab. rob.	FraudBench	Ours
ISO 20022 message elements	no	no	no	yes
Field-placement manipulation	no	no	no	yes
Adaptive or strategic adversary	partial	yes	yes	yes
Explicit ordinary-error floor	partial	no	no	yes (η)
Attacker capability or constraint model	partial	yes	yes	yes
Fixed alert-capacity metric	no	no	no	yes
Evasion-lift metric	no	no	no	SEL

it shares the asymmetric capability framing but operates on dialogue, not on payment messages. Those attacks move numeric feature values under a norm or a constraint set. The manipulations studied here are discrete, semantics-preserving edits to a standardised message, whose cost is operational rather than geometric, and the optimisation follows the classical cost-bounded evasion setting [22], [23]. Because ordinary defects and deliberate edits leave similar traces, the benchmark treats the defect rate η as an explicit parameter rather than a fixed property of the data. Table I summarises the scope of these adjacent resources. Numerical comparison across them would be meaningless because the tasks and data differ; the comparison that can be made is one of scope. The literature reviewed here does not provide a benchmark that combines message-level field operations with a controlled accidental-error floor. This is a scoped observation, not a claim of exhaustive priority.

III. THREAT MODEL AND MANIPULATIONS

The adversary can construct or amend a payment instruction but does not control a bank’s screening system, sanctions lists, or alert threshold. This covers a corporate originator, a payment-service provider (PSP), or an insider with legitimate initiation access. Message-path substitution is a stronger capability: it applies only to a PSP, bank-side actor, or originator that can choose among available channels. A typical corporate sender cannot simply select an interbank message type.

A black-box adversary knows public message guidance and receives only a coarse accept-or-hold outcome; its admissible set excludes the targeted code substitutions P2a, P2b, and P2d, which require knowing which codes score lower, but retains code omission (P2c). A grey-box adversary also knows the relative ordering of the benchmark’s purpose-code weights and may use every primitive. Neither version observes the exact threshold or a production rule set. The distinction is methodological: it lets the evaluation ask whether more knowledge makes the evader harder to catch, without suggesting that any particular bank exposes those weights.

Let x be an illicit synthetic payment instruction, m an admissible modification, $c(m)$ its operational cost, and $\pi(x)$ the alert probability in the published screening abstraction. The

TABLE II

THE ELEVEN BENCHMARK PRIMITIVES WITH THEIR OPERATIONAL COST $c(m)$. COSTS ARE ILLUSTRATIVE CONSTANTS ON A COMMON SCALE; THEY ARE PUBLISHED WITH THE CODE AND ARE NOT ESTIMATES OF ANY ACTOR’S EFFORT.

ID	Operation	$c(m)$
<i>P1 misfielding: where identity or address evidence appears</i>		
P1a	move the ultimate-party name into unstructured remittance text	0.30
P1b	move town and country into a free-text address line	0.25
P1c	truncate a screened name to four characters	0.15
P1d	omit the ultimate debtor	0.20
<i>P2 code choice: declared semantics of an intact container</i>		
P2a	replace the purpose code by the lowest-weight plausible code	0.05
P2b	replace the category purpose	0.08
P2c	omit the purpose code	0.04
P2d	declare a salary purpose over a goods or invoice narrative	0.10
<i>P3 structure: message path or decomposition</i>		
P3a	switch between pain.001 and pacs.008	0.45
P3b	multiply the in-file split count by 3 to 7, capped at 200	0.35
P3c	insert an intermediary as the ultimate party	0.55

evader chooses

$$m^* = \arg \min_m \{ \pi(m(x)) + \lambda c(m) \} \quad \text{s.t.} \quad V(m(x)), |m| \leq k, \quad (1)$$

where V requires the modified record to pass the benchmark’s structural and business checks and preserve the intended transfer. V holds by construction: every primitive rewrites placement, codes, or path while leaving the amount, the parties, and the intended beneficiary unchanged, the primitives of a subset are applied in the order of Table II, and no pair is mutually exclusive. This is a cost-bounded, test-time evasion problem [22], [23]. It does *not* imply that every generated XML instance would be accepted or settled by a production bank.

Table II lists the primitives. P1 changes where identity or address evidence appears. P2 leaves the container intact but changes the declared semantics. P3 changes the message path or decomposition. These families describe the adversary’s action. Whether they also predict detectability is treated as an empirical question; the evaluation tests that expectation rather than building it into the detector.

IV. BENCHMARK AND METHOD

A. Synthetic population and edits

The generator draws each message independently. A creditor belongs to one of six industries, each with a fixed purpose-code profile over eight purpose codes (for example, a manufacturing creditor receives GDDS with probability 0.45, SUPP 0.30, TRAD 0.15, SERV 0.05, and OTHR 0.05). A debtor-creditor corridor is drawn from five country pairs with fixed corridor weights, the category purpose from five codes, the amount from a log-normal distribution with log-mean 10.2 and log-standard-deviation 1.4, and the message type from pain.001 and pacs.008 with equal probability. An ultimate debtor is present with probability 0.75, and a pair identifier drawn from $n/20$ values links messages that share a synthetic debtor-creditor relationship, so that a 200,000-record run has about twenty messages per pair. Party names are 4 to 16 random letters. Unstructured remittance text is an invoice reference followed by

zero to six words from a small vocabulary and, with probability 0.3, a second reference, so benign lengths spread over a range instead of taking a few fixed values. The three words that C4 keys on are not in that vocabulary, since C4 is defined by them. Pain.001 files may contain ordinary batches, so that multiple transactions are not a perfect gaming indicator: 30% of benign pain.001 files are batched, nine in ten of them with 2 to 49 transactions and the rest with 50 to 200.

A fraction η of records receives one ordinary data-quality defect. The default mix is 35% purpose content placed in remittance text, 30% partial address destructuring, 20% absent ultimate party, and 15% absent category purpose. These proportions are controlled benchmark choices, not measured industry rates. Their classes follow the types of field-use problems described by PMPG [4].

A fraction ρ is then marked illicit for the screening abstraction. For each illicit record the evader enumerates every subset of admissible primitives of size at most k , applies the subset, and keeps it if $\pi(m(x)) + \lambda c(m)$ falls below the unmodified $\pi(x)$. A perfectly rational minimum would concentrate on one cheap edit and leave too few samples for several primitives, so the implementation samples among the kept subsets and the no-edit option with probability proportional to $\exp(-(\text{objective} - \text{minimum})/T)$. Illicit records for which the no-edit option is drawn form the unevaded control group. Ground truth records the chosen primitives, the alert probability before and after the edit, and any accidental defect already present.

B. Screening abstraction and calibration

The incumbent alert model is transparent rather than institution-specific. A name leg matches listed names against a stated set of screened elements (party names, town, and country); address lines and remittance text are not screened, which is the coverage gap P1 exploits. For an illicit record whose listed name sits in a screened element, the name leg alerts with probability P_{name} , reduced to $0.39P_{\text{name}}$ when the name has been truncated below a fuzzy-match length. A score leg combines purpose, corridor, amount, and completeness,

$$s(x) = \frac{w_p + w_c + 0.2 \min(1, a/250,000) + 0.10 u + 0.08 t}{\sqrt{n_{\text{splits}}}}, \quad (2)$$

where a is the amount, u indicates a missing ultimate debtor, and t a missing town or country, followed by a logistic threshold, $p_{\text{score}} = 1/(1 + e^{-8(s-\tau)})$. The purpose weight w_p runs from 0.08 (SALA) to 0.45 (TRAD), with 0.32 for an absent code; the corridor weight w_c runs from 0.05 to 0.40. Splitting a file dilutes the amount term, which is the mechanism behind P3b. A residual floor f represents controls outside the evader’s reach and applies to illicit records only. The combined probability is

$$\pi(x) = 1 - (1 - p_{\text{name}})(1 - p_{\text{score}})(1 - f). \quad (3)$$

For benign records only the score leg is active. The weights are illustrative and published with the code. They are not estimates of any bank’s rule set.

TABLE III

CONSTRAINT GROUPS AND THEIR FEATURE DEFINITIONS. q_{ind} IS THE CREDITOR INDUSTRY’S PROFILE PROBABILITY OF THE DECLARED PURPOSE CODE (0 WHEN ABSENT).

ID	Feature	Context
C1	1 if the declared purpose code is not in the code list	field
C2	[TwnNm absent] + [Ctry absent] + [an AdrLine carries a town or country token], each 0 or 1	message
C3	$(1 - q_{\text{ind}}) + [\text{UltmtDbtr absent}]$	message
C4	1 if the purpose is SALA and the remittance text mentions goods, invoice, or shipment	message
C5	share of the pair’s other-fold messages that declared a different purpose	history

Calibration deserves explicit treatment because P1 mainly attacks the name leg and P2 mainly attacks the score leg. Setting the desired baseline alert probability π_0 by adjusting only one leg can make the other family irrelevant. The benchmark instead divides baseline survival multiplicatively between the two legs using a stated name-share parameter σ . With $S = (1 - \pi_0)/(1 - f)$, the name leg is set to $P_{\text{name}} = 1 - S^\sigma$ directly. τ is then solved by bisection so that the mean score-leg probability over an unevaded illicit probe sample of 4,000 records equals $1 - S^{1-\sigma}$. The main setting uses $\pi_0 = 0.45$ and $\sigma = 0.6$, which solves to $P_{\text{name}} = 0.256$ and $\tau = 0.768$; a 3×3 grid varies $\pi_0 \in \{0.30, 0.45, 0.60\}$ and $\sigma \in \{0.4, 0.6, 0.8\}$.

C. Validation features and XML

Five constraint groups produce detector features. Table III gives their definitions and separates local checks from those needing context. C1–C2 represent low-cost validation; C3–C4 look for contradictions within a message; C5 requires pair history. The purpose-history feature reflects the type of new-code and historical-pattern mismatch described by the Federal Reserve [24]. C5 compares a message’s declared purpose with the purposes declared by the same pair’s messages in the other cross-fitting fold, gamed or not; no gaming label enters any feature, and a message does not contribute to the history used to score it. Six auxiliary features complete the eleven-dimensional vector: purpose absent, remittance length, number of address lines, split count, ultimate debtor absent, and a creditor name of at most five characters. The supervised detectors use a separate 25-dimensional raw vector: one-hot purpose, category purpose, and corridor (each with an “absent” slot), log amount, split count, ultimate debtor absent, town absent, and remittance length. The raw vector deliberately excludes the evader’s own objective terms.

The field-level records can be serialised as pain.001.001.09 or pacs.008.001.08. The release checks XML well-formedness and exposes an optional XSD-validation hook when schemas are supplied. No XSD-validation outcome is used as experimental evidence, and the paper therefore does not claim that all emitted instances are production-valid. This distinction also avoids conflating an XSD pass with channel rules, business validation, or settlement.

D. Detectors, reference methods, and cross-fitting

$D_0 = C1 + \mathbb{I}[C2 > 1]$ applies the cheap checks. $D_1 = C1 + 0.5 C2 + C3 + 1.5 C4$ adds the fixed consistency rules with hand-set weights. D_2 is a histogram gradient-boosted classifier (150 boosting iterations, other settings at the scikit-learn defaults) trained on the raw vector with gaming labels. D_3 is an isolation forest with 200 trees fitted to the eleven-dimensional feature vector of benign training rows [25]. D_4 is the per-run sum $z(D_1) + z(D_3)$; it has no learned fusion weight and uses no gaming labels. Two standard reference methods are added so that the benchmark’s detectors can be compared with existing techniques on identical inputs. D_5 is a local outlier factor with 20 neighbours [26], fitted in novelty mode to the standardised benign feature vector. D_6 is an ℓ_2 logistic regression on the standardised raw vector with gaming labels. No class weighting is used, and no hyperparameter is tuned on the benchmark, so no validation split is needed. Only D_2 and D_6 train on gaming labels. The label-free detectors D_3 and D_5 still assume a trusted benign training stream: they use no positive label, but they do rely on knowing that their training rows are unmanipulated.

Every learned model is two-fold cross-fitted. Records are split by index parity into two folds; each model is fitted on one fold, restricted to the rows it is allowed to see, and scores only the other fold. Every reported score is therefore out-of-fold, and no record is scored by a model that trained on it. The same fold split serves two further purposes. A record’s C5 history is built from its pair’s messages in the other fold only, and the standardisation statistics behind D_4 are taken from the other fold’s scores. A record therefore contributes neither to the history nor to the normalisation used to score it. D_0 and D_1 need no fitting. The held-out-family experiment uses the same split, with the training fold additionally excluding every record touching the withheld family.

E. Evasion metrics

AUC measures ranking but not the value of an attack. Let π_0 be the alert probability before manipulation and π_m after it. We define

$$\text{SEL}(m) = \frac{1 - \pi_m}{1 - \pi_0}. \quad (4)$$

If detector D alerts manipulated and unmanipulated illicit payments at rates δ_m and δ_0 , the residual lift is

$$\text{SEL}_{\text{res}}(m, D) = \frac{(1 - \pi_m)(1 - \delta_m)}{(1 - \pi_0)(1 - \delta_0)}. \quad (5)$$

SEL of 1 means that the edit provides no modelled benefit. The gap between SEL and residual SEL is the portion removed by the detector.

F. Evaluation protocol

The main setting uses $n = 200,000$ records per seed, seeds 7–9, $\eta = 0.15$, $\rho = 0.05$, $k = 2$, $\lambda = 0.18$, $T = 0.10$, $f = 0.10$, target $\pi_0 = 0.45$, and $\sigma = 0.6$. The positive class is the set of gamed records. About 99.4% of illicit records select at least one edit, so gamed records make up 5.0% of each run

(9,795–10,050 per seed). The negative class is everything else: benign records with or without an accidental defect, and the 53–61 unevaded illicit records per seed. We report AUC with a DeLong 95% interval, average precision (the area under the precision–recall curve), and the true-positive rate (recovery) at exact alert capacities of 1% and 0.1%, each with a 95% record-bootstrap interval from 200 resamples in which the threshold is re-solved. Precision at 1% capacity is recovery divided by the capacity ceiling, since the number of alerts is fixed. Because prevalence is about 5%, no detector can recover more than capacity divided by prevalence: 20.2% of gamed records at 1% capacity and 2.0% at 0.1%. We state these ceilings beside the results.

Discrete detector scores create ties. A plain “score above quantile” rule can therefore alert far more records than the stated capacity. The implementation selects every score above the boundary and assigns the tied block the label-blind fraction needed to fill the capacity exactly. At 1% of 200,000 records this is 2,000 expected alerts, not all records tied with the 2,000th. Reported recovery is the expectation under uniform tie-breaking.

For generalisation, the supervised detectors are retrained while one primitive family is withheld. The evaluation uses $k = 1$ and pure-family messages, preventing a held-out family from leaking through a second primitive. Each held-out AUC is compared with an in-distribution AUC on the same rows from a model trained on every family; label-free detectors have no gap by construction and are shown once. Before the final rerun, the analysis plan recorded two directional expectations: P1 and P3 would behave similarly as coverage changes, while P2 would be harder for fixed rules. A family AUC near chance or SEL near 1 counted against that framing. This was a documented analysis plan, not an external preregistration. The protocol also varies η , k , adversary knowledge, and the calibration grid, each on 100,000 records at seed 7. Configuration, seeds, the commit of the harness, and raw JSON outputs are retained with the release; the numbers reported here were produced at commit 378369dbe6e0.

V. RESULTS

A. Ranking versus recovery at capacity

Table IV(a) separates ranking from useful recovery. At 1% capacity, 2,000 alerts must cover about 9,910 gamed records, so no detector can recover more than 20.2%. Among the label-free detectors the ordering by AUC ($D_5 > D_4 > D_1$) is the reverse of the ordering by recovery. D_5 has the best AUC (0.777) and recovers 3.8%; the fixed rules D_1 have the lowest AUC of the three (0.750) and recover 10.1%, half of the ceiling, and at 0.1% capacity they fill all 200 alerts with gamed records. The precision column shows the difference: half of D_1 ’s alerts at 1% are gamed records, against 27% for D_4 and 19% for D_5 . A score that is high only when a message contradicts itself places those records ahead of everything else, while the density and forest scores also rank rare benign combinations highly. Average precision, which weights the top of the ranking, puts D_1 first among the label-free detectors as recovery does;

TABLE IV
DETECTION RESULTS. PANEL (A) IS THE MEAN $\pm$ STANDARD DEVIATION OVER THREE 200,000-RECORD RUNS; RECOVERY (REC.) USES EXACT CAPACITY WITH LABEL-BLIND TIE-BREAKING, PRECISION (PREC.) IS RECOVERY DIVIDED BY THE CEILING, AND THE LAST ROW IS THE CEILING IMPOSED BY CAPACITY AND PREVALENCE. PANEL (B) IS THE MEAN OF THREE PURE-FAMILY RUNS AT $k = 1$; n IS ROUNDED, AND ONLY D_2 AND D_6 ARE RETRAINED. $\dagger$ TRAINS ON GAMING LABELS.

(a) Main setting						
Detector	AUC	AP	Rec.@1%	Prec.@1%	Rec.@0.1%	
D_0 C1–C2 checks	0.497±0.001	0.049	0.009±0.000	0.043	0.001	
D_1 fixed rules	0.750±0.004	0.271	0.101±0.003	0.500	0.020	
D_3 isolation forest	0.699±0.007	0.097	0.032±0.003	0.160	0.005	
D_4 $z(D_1) + z(D_3)$	0.762±0.005	0.168	0.055±0.003	0.271	0.011	
D_5 LOF	0.777±0.002	0.244	0.038±0.004	0.189	0.014	
D_2 gradient boosting [†]	0.883±0.002	0.499	0.168±0.001	0.832	0.020	
D_6 logistic regression [†]	0.784±0.004	0.313	0.127±0.001	0.628	0.018	
Capacity ceiling	–	–	0.202	1.000	0.020	
(b) Leave-one-family-out						
Family	n	D_1	D_4	D_5	D_2 in $\rightarrow$ out	D_6 in $\rightarrow$ out
P1 misfielding	3970	0.736	0.781	0.566	0.769 $\rightarrow$ 0.377	0.751 $\rightarrow$ 0.368
P2 code choice	1777	0.848	0.762	0.823	0.958 $\rightarrow$ 0.617	0.950 $\rightarrow$ 0.668
P3 structure	3872	0.448	0.460	0.594	0.627 $\rightarrow$ 0.473	0.466 $\rightarrow$ 0.460

AUC puts it last. The supervised D_2 reaches AUC 0.883 and recovers 16.8% (83% of the ceiling, precision 0.832), but it assumes labels that are unlikely to exist for new manipulations. Three-seed standard deviations are at most 0.007 for AUC and 0.004 for recovery, and the widest DeLong half-interval is 0.006, so each difference discussed here exceeds its uncertainty.

B. Comparison with reference methods

The two reference methods bracket the benchmark’s own detectors. The local outlier factor D_5 , given the same eleven features as D_3 , beats the isolation forest on every metric (AUC 0.777 against 0.699) and edges the rule-and-forest combination D_4 on AUC, yet recovers less at 1% capacity. Logistic regression D_6 on the same raw features as D_2 reaches AUC 0.784 and recovers 12.7%, 2.6 points above the best label-free detector; the boosted model adds a further 4.1 points. Labels help, and at this capacity the nonlinear model is worth as much again. Neither reference method changes the qualitative picture: label-free detection recovers a minority of the capacity ceiling, and supervised detection recovers most of it only in-distribution.

Because the generator writes the features the detectors read, we audited whether any detector’s alerts rest on values that benign synthetic traffic does not produce. In the main setting 1.1% of gamed records carry a remittance length or split count outside the benign range, no benign record does, and at most 5% of any detector’s alert mass at 1% capacity falls on such records (seed 7). An earlier generator version emitted benign remittance text of exactly two lengths; every primitive that appended text then left a length absent from benign traffic, and a one-line out-of-range rule recovered the entire 1% capacity with no false alert. The audit is kept in the release as a regression check, and the numbers reported here are from the corrected generator.

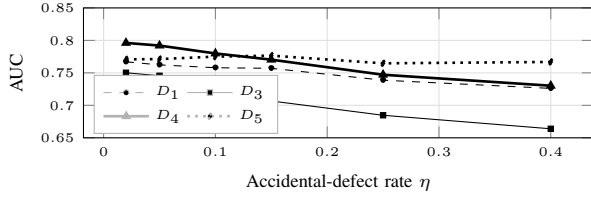


Fig. 1. AUC as the ordinary defect rate rises. Each point uses 100,000 records at seed 7; only AUC, which is unaffected by threshold ties, is shown.

C. Generalisation to an unseen family

Table IV(b) shows how brittle the supervised advantage is. With P1 withheld, D_2 falls from 0.769 to 0.377 on the same rows (per-seed 0.370–0.382), and D_6 falls from 0.751 to 0.368. The P2 drop is smaller but still material (0.958 to 0.617). P3 is not separated by any message-level detector: D_1 scores 0.448, D_4 0.460, and the withheld D_2 0.473. This does not prove that P3 is inherently hidden; it shows that the current fields offer little ranking signal.

A held-out AUC well below 0.5 is a systematic inversion, not an absence of signal, and its mechanism can be read off the training data. With P1 withheld, an absent ultimate debtor is more common among benign training records (27%, the generation rate plus the accidental-omission defect) than among the remaining gamed ones. The gamed rate given an absent ultimate debtor is 2.0%, against 3.2% when it is present (seed 7, 100,000 records, $k = 1$). The model learns absence as evidence of benignity. P1a and P1d both remove the ultimate debtor, and their held-out AUC against benign traffic is 0.313 and 0.296; P1c, which truncates a name and leaves the ultimate debtor in place 81% of the time, scores 0.570. The ordinary-error floor therefore does more than add noise: it teaches a supervised model that the signature of an unseen misfielding edit is what routine sloppiness looks like. P1b is rarely selected on its own (4 pure selections in 100,000 records at $k = 1$) because the screening abstraction has no address-based term, so the P1 rows are carried by P1a, P1c, and P1d.

D. Noise floor

Fig. 1 shows gradual degradation rather than collapse. Raising η from 0.02 to 0.40 lowers D_4 AUC by 0.066 and D_3 by 0.086; the fixed rules lose 0.041 and are the most stable of the benchmark’s detectors. The anomaly model degrades fastest because the benign distribution it learns is itself moving. D_5 moves by no more than 0.012 across the sweep; it keys on rare combinations rather than on how common the ordinary defects are. At the highest noise rate every detector still ranks above chance, but the main operational result warns against treating that ranking as sufficient.

E. Screening-evasion lift

Manipulation increases modelled pass-through by 33%–79% in the main calibration. Per-seed 95% bootstrap half-widths from 1,000 resamples are at most 0.025 for SEL and 0.033 for residual SEL, conditional on the observed unevaded baseline.

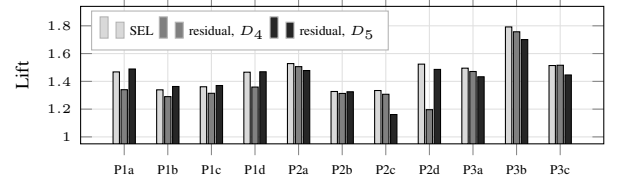


Fig. 2. Mean screening-evasion lift before and after detection at the 1% capacity, three seeds. Values are relative to the published screening abstraction.

TABLE V
 D_4 AND D_5 ABLATIONS, 100,000 RECORDS AT SEED 7 PER SETTING.

Factor	Setting	D_4 AUC	D_5 AUC
Family	P1 / P2 / P3	0.843 / 0.799 / 0.488	0.668 / 0.863 / 0.728
Budget k	1 / 2 / 3	0.649 / 0.770 / 0.844	0.614 / 0.776 / 0.845
Knowledge	black / grey	0.728 / 0.770	0.747 / 0.776

The detector’s value varies sharply by primitive and by detector. For the code–narrative mismatch P2d, D_4 alerts 22.0% of instances and cuts the lift from 1.524 to 1.196; the residual stays above 1, so the check removes part of the benefit at this capacity, not all of it. D_5 barely touches P2d (4.2%) but is the one of the two that reaches the omitted-code edit P2c (14.5% against 2.6%). For in-message splitting P3b the lift is the largest of all (1.792) and changes little under either detector (1.757 and 1.701). P3c is starker still in detection rate: D_4 alerts 0.5% of its instances. At $k = 2$ most gamed messages carry two primitives, so a per-primitive rate counts a message whichever edit triggered the alert.

Across the three seeds, D_4 alerts 0/61, 1/59, 0/53 unevaded illicit records, so the observed δ_0 is near zero. Replacing each seed’s δ_0 by its exact two-sided 95% upper bound would raise every residual value by at most 8% and changes no ordering; the small control group remains a reason not to over-interpret exact values. Across the 3×3 calibration grid, SEL ranges from 1.13 to 2.56 while D_4 AUC stays between 0.749 and 0.802. The size of the modelled threat is therefore sensitive to the assumed baseline and name share, while the relative detection difficulty is more stable. This is why the release reports the grid rather than treating one SEL number as a deployment estimate.

F. Calibration grid and ablations

Table V supports the same interpretation. More edits make the evader easier to rank: D_4 AUC rises from 0.649 at $k = 1$ to 0.844 at $k = 3$, because each edit can leave another inconsistency. The grey-box adversary is easier for D_4 to detect than the black-box one (0.770 against 0.728): acting on code-weight knowledge touches fields that C3–C4 examine. A more informed adversary is therefore not automatically a stealthier one in this benchmark. Family-only runs put P3 far below P1 and P2 for D_4 . For D_5 the hardest family is P1 (0.668 against 0.843 for D_4), which matches its weak held-out P1 score in Table IV(b); D_5 is the stronger of the two on P2 and P3.

G. What predicted detectability

The family-level expectations did not survive the rerun. P1 and P3 were both treated as coverage changes, yet D_1 scores 0.736 on P1 and 0.448 on P3. P2, expected to be harder for fixed rules, is D_1 's easiest family at 0.848. The family taxonomy therefore organises the edits but does not explain the rankings. The sharper distinction is whether an edit leaves a checkable contradiction. Within P2, D_4 alerts 22.0% of P2d instances at the 1% capacity but only 1.6%–2.6% of P2a–P2c. Several P1 edits leave placement or completeness clues, whereas path and decomposition changes often remain internally coherent. We retain P1/P2/P3 as an action taxonomy, not a detectability taxonomy.

VI. DISCUSSION AND LIMITATIONS

A. Design implications

The practical lesson is not that structured data weakens controls. It gives controls better inputs and enables useful consistency checks. The narrower lesson is that validation, screening coverage, and behavioural monitoring must be designed together. C1–C2 catch malformed or incomplete data. C3–C4 catch contradictions within one message. C5 or other longitudinal features are needed when the message is internally plausible. Changes to path, batch structure, or party chain may require channel history, agent-chain context, or comparison with the originator's normal files. A message-only model cannot infer evidence that the message does not contain. That distinction is the study's main empirical finding.

The study also separates detector credit from incumbent-control credit. A primitive can be easy to classify yet offer little extra protection if the incumbent model would already alert it. Conversely, modest AUC can still be useful for a narrow primitive if it removes substantial lift at the operating threshold. SEL and residual SEL make that distinction visible, but they inherit the assumptions of π .

B. Limitations

Several limitations remain. The screening abstraction is transparent and reproducible, not a reconstruction of a bank's engine, so SEL is a stress measure rather than an estimate of losses or real evasion. The distributions of purpose codes, defects, batching, and illicit traffic are assumptions. The generator and detectors share a feature vocabulary, which can reward recognition of implementation artefacts; the leave-one-family-out collapse of the supervised models demonstrates the risk rather than solving it. The feature vector is discrete, so many records coincide exactly, and neighbourhood methods such as D_5 rank rare combinations rather than smooth densities; its ranking is therefore sensitive to how the generator populates rare cells, which the artefact audit of Section V-B is designed to expose. Cross-fitting is done within one corpus by index parity, not by time, so it removes memorisation but does not test drift. External validity requires recalibration on an institution's own traffic.

The XML layer has a similarly narrow claim. The serializers are checked for well-formedness, the pain.001 ultimate-debtor

ordering was corrected during review, and the harness accepts external XSDs. The release does not treat XSD validity as proof of business acceptance. P3a is also capability-dependent and should not be attributed to an ordinary corporate originator without route choice. A stronger next experiment would calibrate π against anonymised alert outcomes inside one institution and evaluate path-aware controls without exporting raw messages.

C. Ethics and responsible release

The paper enumerates operations that lower a modelled alert probability, so the release decision deserves a direct answer. None of the eleven operations is disclosed here for the first time: registered code lists, the harmonised data requirements [2], and the published guidance on which elements normally warrant direct screening [3] are public, and market-practice guidance already records purpose codes being placed in the wrong field [4]. What this work adds is measurement, not capability. The asymmetry favours the defender: a motivated adversary can learn which fields an institution reads by observing its own payments, whereas a control owner has had no structured way to enumerate the coverage gaps that follow from a selective screening policy. An institution can map the families onto its own screened fields, message paths, and history features without disclosing any payment data. The release is deliberately non-operational. The purpose, corridor, and cost weights are illustrative constants; none is calibrated against a real corridor, code list, or deployment, and the corpus contains no customer records, employer data, or production alert outcomes.

D. Artifact availability

The generator, the evaluation harness, the configuration files, the seeds, and the raw per-run JSON outputs behind every table and figure are released, with a notice that the corpora are synthetic and must not be represented as real payment traffic. Every result file records the commit of the harness that produced it (378369dbe6e0 for the results reported here). A machine-readable summary collects every number used here, and a verification script re-derives each of them from the raw outputs. Each number can therefore be reproduced from its stated seed and configuration, not from a rebuilt approximation. The release is public at <https://github.com/abhisheksharma2411/ISO-20022-purpose-code-gaming-benchmark> under the MIT licence and archived at Zenodo under DOI 10.5281/zenodo.23049366.

VII. CONCLUSION

Some message gaming remains distinguishable from routine error in this benchmark, and some does not. Edits that leave a contradiction inside the message, such as a salary code over a goods narrative, are caught by a fixed consistency rule. Edits that remove or reroute information while leaving the message coherent, such as path substitution or file splitting, are not separated by any message-level detector tested here. Good AUC does not imply useful recovery at a small alert capacity. The label-free detectors reach AUC 0.750 to 0.777 yet recover

3.8% to 10.1% of gamed messages when 1% of traffic may be alerted, against a ceiling of 20.2%, and the best-ranking one recovers the least. A supervised model closes most of that gap in-distribution and loses it on a family it has not seen, because the ordinary-error floor makes the unseen family’s signature look benign. The release provides a compact way to test that boundary and to challenge its assumptions.

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